

Cultivating Change for Land Degradation Neutrality through Evidence-Based Policy and Practice – The APCMNF Natural Farming Model”



Andhra Pradesh Community Managed Natural Farming (APCMNF)
Vision 2031 - all 6.0 million farmer households, and 2.0 million farm worker households



Soil Degradation And Achieving Land Degradation Neutrality

Land degradation – the loss of land's health

Land degradation continues to impact **3.2 billion people** negatively worldwide

90 percent of land will bear our imprint by **2050**

32 Gigatons of the carbon will be lost from the land b/w 2015-

30 due to the current farming practices

3 concurrent actions to achieve LND Goals

1. avoiding new degradation of land
2. Reducing existing degradation
3. Ramping up efforts to restore and return degraded lands

Farmer Distress Food system – multiple crises



High Costs of Cultivation (Seeds, Fertilizers,

Prolonged Dry Spells, Droughts

Acute water shortages, Drying of Borewells

More frequent cyclones, floods, unseasonal rains

Problems of Small, marginal farmers and Tenants, Rural-Urban

Market Uncertainty

Consumer Food Plate



Food Scarcity

Chemical Residues

Heavy metal contamination in food

Lack of micro nutrients, trace minerals

Soil degradation, Continuous loss of soil organic matter

Water stress and water emergency

Heatwaves – global warming

Decreased bio diversity
Water and air pollution

Environment crisis



Climate emergency

APCNF programme – People's Movement for secure future



86% - small and marginal farmers.

**40,656 farmers
704 (v)
2016-17**

**480,000 farmers
3730 (v)
220,000 Ha
2020-21**

**27% of villages
12% of farmers
4.8 % of area**

**630,000 farmers
3730 (v)
290,000 Ha
2021-22**

**27 % of villages
14 % of farmers
6.3 % of area**

**851,000 farmers
3730 (v)
378,000 Ha
2022-23**

Target

**31 % villages
20 % of farmers
8.3 % of area**

**1,200,000 farmers
4260 villages
500,000 Ha
2023-24**

Largest Natural farming programme in the country, in terms of farmers enrolled.

**Funds: Govt – PKVY, NMNF, KfW loan
Grants: 1. Azim Premji Philanthropy
2. Co Impact**

**Transition of a farmer – 3 to 5 years
No cash incentives during transition, and, no promises of market premia after transition**

Women in Natural Farming: Our biggest Strength



Programme
Management,
transparency

Collective
Action

Peer

7550 village SHG federations, **202,000** women SHGs with
a membership of **1,880,000** women are in charge

Farming
Plans, and,
consumption
plans

Inclusive of
the poorest



APCNF Implementation – overcoming obstacles through critical innovations



Government
support and
advocacy –
resources and
implementation

Knowledge –
POPs, videos, etc
Research
Innovations and
continuous learning

Social capital -
Women SHG s
and federations

**Unique innovations of Govt of A.P and pro farmer policies
and welfare measures, across the value chain.**

**Rythu Bharosa Kendram – farmer service centres – one
stop shop**

Human capital
Farmer to farmer
– extension
system, Knowledge
intensive

Facilitating
organizations –
Govt., NGOs
and C.B.Os

Collaborations with
Global and National
institutions and
Scientific experts

Universal Principles of Natural farming



**Mimicking
Nature**

Soil to be covered with crops 365 days, (Living root principle)

Diverse crops , 15 – 20 crops, include trees

Keep soil covered with crop residues, whenever living plants are not

there Minimal disturbance of soils – minimize tillage

Farmers' own seeds to be used. Indigenous seeds preferred

Integrate animals into farming

Bio stimulants as catalysts to trigger soil biology

Pest management through better agronomical practices and botanical pesticides

No synthetic fertilizers pesticides herbicides weedicides

Microbial seed coating Beejamrutham



Ingredients

Cow dung
– 2
kg

Cow urine
– 2
liters

Lime –
40
grams

Handful of
chemical
free soil

Water –
20
liters



Step 1

Wrap the cow dung in a cloth and submerge in water and let it soak for 12 hours

Squeeze the cloth after 12 hours, add lime, chemical free soil. Mix well in clockwise direction



Step 2

Spray the concoction on all seeds and ensuring each seed is coated by it before sowing



Step 3

Soil Microbial enhancement – Bio stimulant - Ghanajeevamrutham



Ingredient

Cow dung -
100
kg

Jaggery -
1Kg

Pulse flour-
1 kg

Cow urine
- 10
liters

Hand full
uncontamin
ated soil



Step 1



Mix all the ingredients properly



Step 2

Make cakes and shadow dry for 5 days for fermentation



Step 3

Apply these cakes in the field

Soil microbial enhancement – Liquid biostimulant - Dravajeevamrutham

Ingredient

s

**Cow dung-
100kg**

**Cow urine-
3-6
ltrs**

**Pulse flour- 2
kgs**

**Jaggery – 2
kgs**

**Water- 200
ltrs**

**Hand full of
uncontamina
ted soil**



Step 1

Add all the ingredients and mix them in clock wise direction



Step 2

**Keep it fermented for 5 days. The colour and smell changes.
Keep mixing it in between**



Step 3



Spraying of Dravajeevarutham in the field

Pest management through botanical stimulants



Yellow and blue
sticky traps



Preparation of *kashayams*
(bio-innoculants for pest
management prepared from
local ingredients)

APCNF IMPACTS

Cost of cultivation - significant reduction - NF costs are much lower than non-NF, across all crops

Yield differences are not significant between NF and Non-NF farms

Significant increase in net income for NF farmers

Independent Assessment by I.D.S

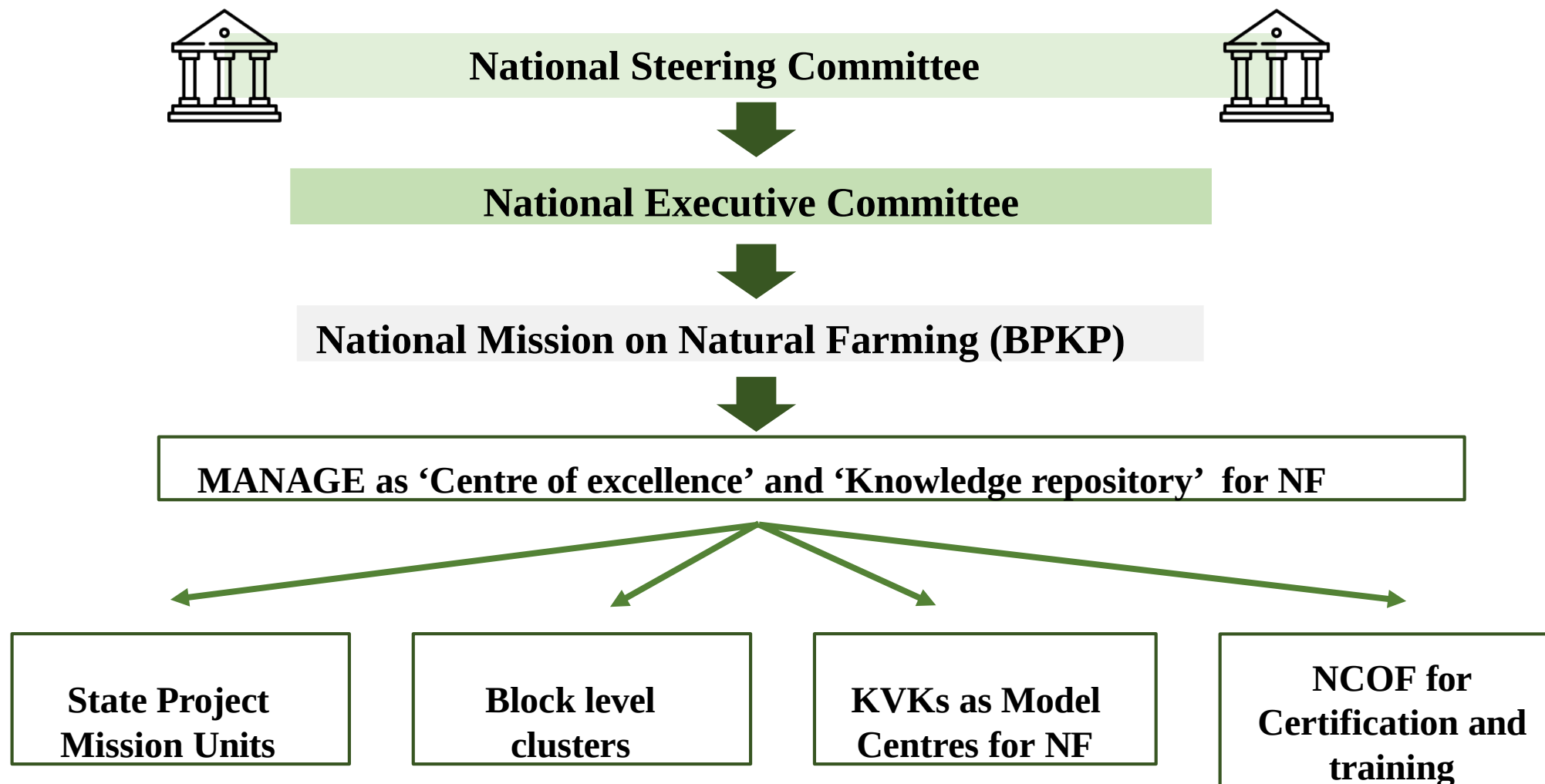
9 seasons till now - 2018-19 (2 seasons) and 2019 – 20 (2 seasons) , Kharif 2020 and Rabi 2020-21 (2 seasons) and Kharif 2021 and Rabi 2021-22 (2 seasons) Rabi 2022-23



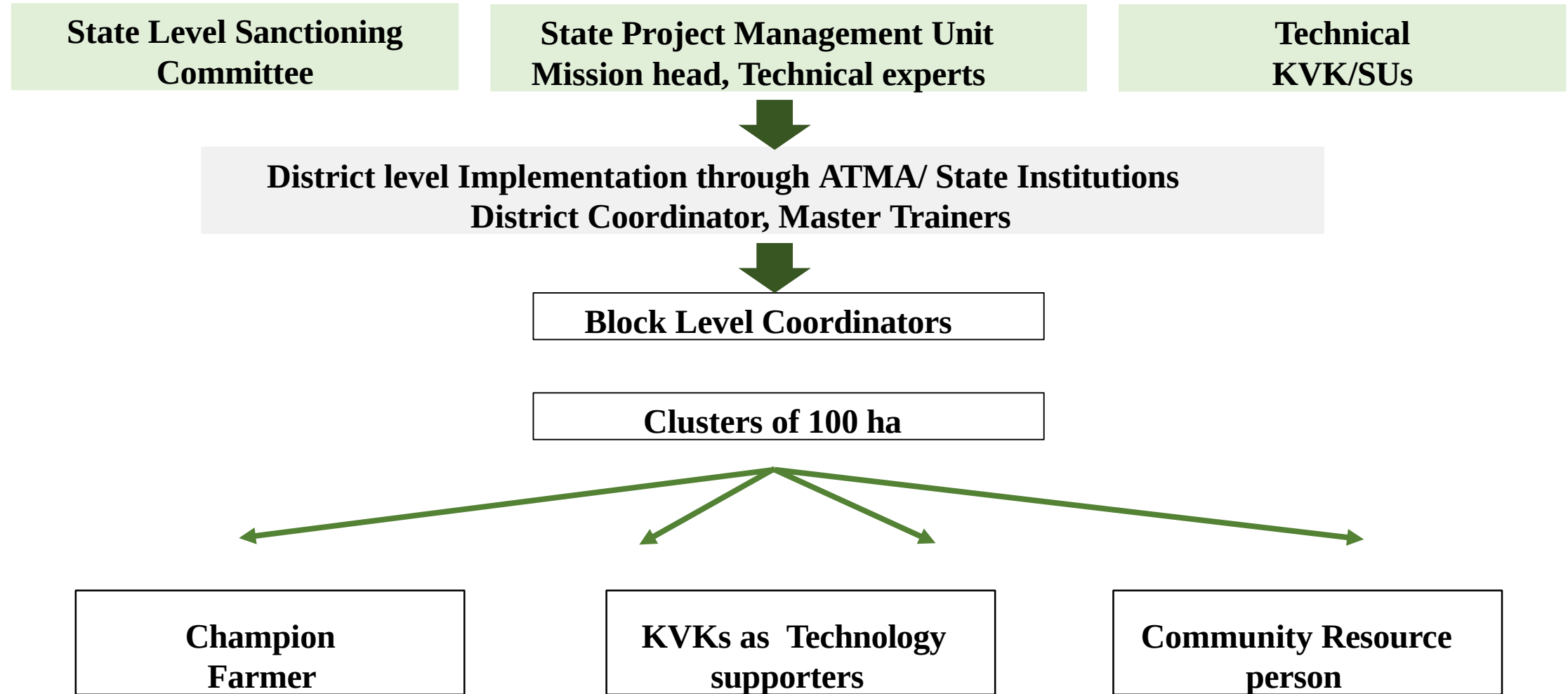
NF farms reported better soil health, crop health, resilience, economic empowerment of farmers and dignity of labor.

The report also mentioned that APCNF has higher potential for expansion of extension services by way of increasing CRPs at the village level

Structured Policy at the National Level



Policy Implementation under Mission Mode (At State Level)



Nation wide Policy to practice and its impact

Paramparagat Krishi Vikas Yojana (PKVY)

- **30,934 clusters (20 ha each) formed since 2015-16**
- **59.75 lakh ha** area and **15.47 lakh** farmers benefitted.
- Area covered NF - **4.09 lakh ha** (980 clusters of 500 ha each in 8 states)
- A dedicated online web portal- www.Jaivikkheti.in

Namami Gange

- 1657 Gram Panchayat in Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal for development of Organic farming in the villages along the river Ganga
- Under **Namami Gange Programme 6,181 clusters and 1,23,620 ha area covered**

Bhartiya Prakrit Krishi Padhti (BPKP)

- GoI is promoting Natural Farming through a sub-scheme named BPKP since 1920-21
- **BPKP same as PKVY with focus on handholding & capacity building**
- [Area covered - 4.09 lakh ha \(980 clusters of 500 ha each\)](#)

Policies at National level

- Clean plant programme
- PM-PRANAM to restore mother earth
- Mission for Integrated Development of Horticulture
- Bio input resource centres



Achievement Evidences of Adopting the policy based practice

NF farmers reported the below environmental impacts noticed after adopting NF practices:

- 64% witnessed increased softness and looseness of soil
- 37% felt crops appeared greener
- 93% saw a decrease in irrigation requirements
- 95% noticed increase in earthworms
- 73% reduced their pesticide usage
- Reaches over **76.21 lakh farmers** in Andhra Pradesh
- Reaches 12 states in India and attracts 45 countries worldwide.
- Prestigious accolades in awards include the **M.S. Swaminathan Award for Environment Protection.**

Gulbenkian award for humanity first time for india - winner 2024



Evidence based policy strategies to achieve LDN

Natural Farming throughout the country NMNF, with a focus on 5-km corridors along river Ganga,

- Clusters to be developed in value chain mode with facilities for postharvest and value addition infra with market linkages
- NCOF to renamed as National Centre for Organic and Natural Farming, NCOF to develop certification through separate portal
- Support to ICAR for developing KVKs as centres of capacity building, training and demonstrations
- **31,500/Ha** for 3 years, **15,000/Ha** period under PKVY, out of this for inputs via DBT

Revision of syllabus *under Agricultural Universities*

- **ICAR committee to draft syllabus and course curriculum for inclusion in UG and PG courses**
- ICAR already notified a dedicated Masters course in Organic farming
- Gujarat Organic Agricultural University (GOAU), Godhra, is running degree courses on natural farming

Way Forward to achieve LDN goals through evidence Based policy and practice - the natural farming way

- Government of India launched the **National Mission on Natural Farming (NMNF)** on **25 Nov 2024**, The scheme has a total outlay of **Rs.2481 crore 2025/26**)
- In next 2 years NMNF will be implemented in **15,000 clusters in Gram Panchayats**, which are willing, & reach 1 crore farmers and initiate Natural Farming (NF) in **7.5 lakh Ha** area.
- need-based **10,000 Bio-input Resource Centres (BRCs)** will be set-up to provide easy availability and accessibility to ready-to-use NF inputs for farmers
- Under NMNF, around **2000 NF Model Demonstration Farms shall be established** at Krishi Vigyan Kendras (KVKs), Agricultural Universities (AUs) and farmers' fields
- **30,000 Krishi Sakhis/ CRPs** will be deployed for awareness generation, mobilisation
- NF could employ twice as many farmers as 10 million by 2050 in AP
- The UNCCD's goal of land degradation neutrality (LDN) can halt, and then reverse, this alarming picture of the future. Already helping 131 of the world's 196 countries that have pledged (or are aiming) to arrest land degradation by 2030.

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“..We do not inherit the earth
from our ancestors, we borrow it
from our children..”

LET US ALL ACT NOW



Thank you for your Attention

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